

Centralized College Merchandise Management System

1 Description of the Use Case

In our college, multiple clubs frequently launch their own merchandise such as T-shirts, hoodies, and other items. Currently, each club manages its merchandise independently, typically through separate forms and size charts. As a result, students face several issues, including repeatedly filling out similar forms, experiencing confusion about varying size formats, and dealing with the absence of a single platform to view all available merchandise.

The proposed system, the **Centralized College Merchandise Management System**, aims to address this problem by acting as a unified portal where all clubs can list and sell their merchandise. Students will be able to create a profile and save their preferred sizes once, eliminating the need to repeatedly enter sizing details for different club merchandise forms.

The primary stakeholders include **students**, who want a streamlined purchasing experience, and **club managers**, who need a structured and efficient way to showcase merchandise and manage student orders.

2 Key Functionalities

Unified Merchandise Catalog

Students can view merchandise from multiple clubs in one place, allowing them to easily browse, compare, and purchase available items.

Saved Size Profile

Students can create a profile and save their preferred size, enabling faster ordering and minimizing confusion about size formats across different clubs.

Delivery & Distribution Updates

Students will receive real-time updates and notifications regarding upcoming delivery and distribution time slots for their orders, ensuring they never miss a pickup.

Club Dashboard for Merchandise Management

A dedicated interface for clubs to publish and showcase merchandise, manage student orders efficiently, and reach a wider student audience.

User Interaction

Users will interact with the system through a **web-based interface** accessible via browsers. Students can browse merchandise, place orders, and receive delivery updates, while club administrators manage merchandise listings and orders through a dashboard.

Technical Highlights

The system will be built using the **MERN stack (MongoDB, Express.js, React, Node.js)**. React will handle the frontend interface, Node.js and Express.js will manage backend APIs, and MongoDB will store application data.

System Architecture

```
Users (Students / Club Admin)
  |
  v
  User Interface
  (Browse, Order, Manage)
  |
  v
  Application Layer
  (Order Management,
  Merchandise Management,
  Delivery Updates)
  |
  v
  Data Layer
  (User, Orders, Clubs,
  Merchandise Data)
```

Design Patterns Used

- **Factory Pattern:** Creates different types of merchandise objects (such as T-shirts, hoodies, caps, or mugs) through a centralized creation method, avoiding direct object instantiation throughout the codebase.
- **Observer Pattern:** Enables real-time updates and notifications. When events such as order confirmation, delivery slot announcements, or distribution updates occur, all subscribed students automatically receive notifications.
- **Publish–Subscribe Pattern:** Implements an event-driven communication system where services publish events (such as delivery schedule updates), and subscribed components like notification services react to those events without tight coupling.
- **Repository Pattern:** Abstracts database operations by providing dedicated repositories for entities such as merchandise, orders, students, delivery slots, and notifications, making the system easier to maintain and modify.
- **Builder Pattern:** Facilitates the step-by-step construction of complex objects such as student profiles that contain multiple attributes like name, contact details, preferred size, and address.
- **Command Pattern:** Encapsulates actions such as placing an order, canceling an order, or updating delivery slots as objects, enabling better management of system operations and task handling.

Architectural Tactics

- **Modularity:** The system will be divided into separate modules such as user management, merchandise management, order processing, and delivery updates to improve maintainability and future extensibility.
- **Scalability:** The architecture will support an increasing number of users, clubs, and merchandise listings without significant performance degradation.
- **Usability:** The platform will provide a simple and intuitive interface with features such as saved size profiles and a centralized catalog to improve the user experience.
- **Security:** Authentication and role-based access control will ensure that only authorized students and club administrators can access relevant system functionalities.
- **Maintainability:** Clear separation of components and use of design patterns will make the system easier to maintain, debug, and extend.

3 Expected Time to Build a Prototype

The project will be developed by a **team of five members over approximately four weeks.**

Week 1 – Research and Planning

Study the current merchandise ordering process in clubs, identify key problems, and define system requirements and core features.

Week 2 – System Design

Design system architecture, database schema, and basic UI wireframes for students and club administrators.

Week 3 – Core Development

Implement merchandise listing, student profile with saved size, and order placement functionality.

Week 4 – Testing and Refinement

Perform system testing, implement delivery slot notifications, fix bugs, and finalize the working prototype.

The prototype will demonstrate three end-to-end functionalities: **centralized merchandise browsing, order placement using saved size profiles, and delivery/distribution update notifications.**

4 Domain

The project belongs to the **E-Commerce / Education Technology domain**. It focuses on improving the process of selling and managing college merchandise by providing a centralized platform for multiple clubs. The system simplifies merchandise browsing, ordering, and delivery updates for students while helping clubs manage orders and distribution more efficiently within the campus ecosystem.